

DATA JOBS MARKET ANALYSIS DASHBOARD

Project Report

Submitted By

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Abstract

The demand for data professionals has increased rapidly due to the growing adoption of data-driven decision-making across industries. Organizations continuously seek professionals with strong analytical and technical skills, making it essential for students and aspiring data analysts to understand current hiring trends.

This project presents a comprehensive analysis of LinkedIn job postings related to data careers. Using Python for data preprocessing and Tableau for visualization, the project examines hiring trends, salary patterns, work arrangements, geographical distribution, and the demand for technical skills.

The analysis was performed on a LinkedIn job posting dataset containing more than 123,000 job listings. After filtering relevant positions, 1,717 data-related job postings were analyzed. The findings revealed significant insights into employer requirements, salary ranges, work types, and hiring locations.

The project demonstrates practical applications of data cleaning, exploratory data analysis (EDA), visualization, and dashboard development. It also highlights the importance of transforming raw data into meaningful business insights that can support career planning and organizational decision-making.

1. Introduction

Data has become one of the most valuable assets for organizations. Businesses across industries rely on data analysts, data scientists, machine learning engineers, and business analysts to extract meaningful insights from large datasets. Consequently, the demand for skilled data professionals continues to grow worldwide.

Understanding the job market is important for students, job seekers, educational institutions, and recruiters. By analyzing job postings, it becomes possible to identify which skills are most in demand, where companies are hiring, what salaries are being offered, and which employment types dominate the market.

This project focuses on analyzing LinkedIn job postings related to data careers. The project uses Python to perform data cleaning, preprocessing, exploratory data analysis, and feature extraction. Tableau is then used to develop an interactive dashboard that presents the findings in a visually appealing and easy-to-understand format.

The overall objective is to transform raw job posting data into actionable insights that help aspiring data professionals understand the current employment landscape.

2. Problem Statement

The rapid growth of data-related careers has created an overwhelming amount of information for job seekers. Thousands of job postings are published every day, making it difficult to understand current market trends.

Without proper analysis, candidates may struggle to answer questions such as:

- Which technical skills are most frequently requested?
- Which job roles offer the highest salaries?
- Which cities have the highest hiring demand?
- What types of employment are most common?
- Which companies are actively recruiting data professionals?

This project addresses these questions by analyzing LinkedIn job posting data and presenting the results through interactive dashboards.

3. Project Objectives

The primary objectives of this project are:

- Analyze LinkedIn job postings related to data careers.
- Identify the most demanded technical skills.
- Analyze salary distributions across different job roles.
- Study geographical hiring trends.

- Understand the distribution of work types such as full-time, contract, internship, and part-time positions.
 - Build an interactive Tableau dashboard to communicate business insights.
 - Demonstrate practical knowledge of Python-based data analysis and business intelligence tools.
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4. Scope of the Project

The scope of this project includes:

- Collecting and analyzing LinkedIn job posting datasets.
- Filtering only data-related job roles.
- Performing exploratory data analysis.
- Extracting meaningful business insights.
- Visualizing the findings using Tableau.
- Creating a professional dashboard suitable for portfolio and interview presentations.

The project does not perform predictive modeling or machine learning. Its primary focus is descriptive analytics and business intelligence.

5. Dataset Description

The project utilizes the **LinkedIn Job Postings (2023–2024)** dataset, which contains job advertisements collected from LinkedIn across multiple industries. The dataset provides detailed information about job postings, companies, salaries, work types, locations, required skills, and industries.

Initially, the dataset contained **123,849 job postings** distributed across multiple CSV files. To focus on data-related careers, the dataset was filtered using keywords such as Data Analyst, Data Scientist, Data Engineer, Business Analyst, Machine Learning Engineer, and Analytics Engineer. After filtering, **1,717 relevant job postings** were selected for detailed analysis.

The primary datasets used in this project are:

postings.csv

Contains job titles, descriptions, company IDs, locations, salary information, work types, and job identifiers.

companies.csv

Contains company names and company-related information.

job_skills.csv

Maps each job posting to one or more skill categories.

skills.csv

Contains skill category information.

industries.csv

Stores industry names.

job_industries.csv

Maps job postings to industries.

The combination of these datasets enabled comprehensive analysis of hiring trends, salary information, and skill demand.

6. Tools and Technologies

The following technologies were used during the development of this project.

Python

Python was used as the primary programming language for data analysis. It was responsible for importing datasets, cleaning missing values, filtering records, performing exploratory data analysis, and exporting processed datasets.

Pandas

The Pandas library was used extensively for data manipulation. It enabled filtering, grouping, aggregation, merging multiple datasets, and calculating statistical summaries.

Matplotlib

Matplotlib was used to create visualizations during the exploratory analysis phase before building the Tableau dashboard.

Tableau

Tableau was used to build an interactive business intelligence dashboard that presents the project findings through KPI cards, bar charts, and interactive visualizations.

Jupyter Notebook

Jupyter Notebook served as the development environment where the complete data analysis workflow was implemented and documented.

7. Methodology

The project followed a structured data analytics workflow consisting of several stages.

Step 1: Data Collection

Multiple CSV files from the LinkedIn Job Postings dataset were imported into Jupyter Notebook using the Pandas library.

Step 2: Data Exploration

The datasets were explored to understand:

- Number of rows and columns
- Data types
- Missing values
- Relationships between datasets
- Key columns used for joins

This step ensured that the structure of the dataset was fully understood before analysis.

Step 3: Data Cleaning

The following preprocessing activities were performed:

- Removed unnecessary records
- Filtered only data-related job postings
- Checked missing values
- Created new summary tables
- Prepared datasets for visualization

Step 4: Exploratory Data Analysis

Several analyses were conducted including:

- Total data jobs
- Company distribution
- Hiring locations
- Salary trends
- Work type distribution

- Technical skill frequency

The objective was to identify patterns and trends within the data.

Step 5: Dashboard Development

The processed datasets were exported as CSV files and imported into Tableau.

Multiple worksheets were created, including:

- KPI Cards
- Top Skills
- Hiring Locations
- Highest Paying Roles
- Work Type Distribution

Finally, these worksheets were combined into an interactive dashboard.

8. Data Analysis

After cleaning and filtering the dataset, several key performance indicators were calculated.

Total Data Jobs

A total of **1,717** data-related job postings were identified from the original dataset.

Companies Hiring

The dataset contained job postings from **1,018 unique companies**, indicating strong demand across organizations of different sizes and industries.

Hiring Locations

The filtered dataset represented **441 unique locations**, demonstrating that opportunities exist across many geographical regions.

Average Salary

The calculated average salary across all analyzed data roles was approximately **\$95,653**, highlighting the strong earning potential of careers in data analytics and data science.

These summary statistics formed the basis for the KPI section of the Tableau dashboard.

9. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to identify important trends before building the dashboard.

Several visualizations were created using Matplotlib to understand:

- Distribution of technical skills
- Hiring locations
- Salary distribution
- Employment types
- Frequency of different job roles

The analysis revealed that full-time employment represented the majority of available opportunities. Senior and machine learning related positions generally offered higher salaries than entry-level roles.

EDA also confirmed that Python, SQL, cloud technologies, and analytical skills appeared frequently across job descriptions, reinforcing their importance for aspiring data professionals.

These findings guided the design of the Tableau dashboard and ensured that the most relevant business insights were presented effectively.

10. Dashboard Explanation

An interactive Tableau dashboard was developed to present the results of the analysis in a visually meaningful and user-friendly manner. The dashboard summarizes important metrics and trends identified during exploratory data analysis.

The dashboard contains the following components:

10.1 KPI Cards

The top section of the dashboard displays four Key Performance Indicators (KPIs):

- Total Data Jobs: **1,717**
- Unique Companies: **1,018**
- Unique Locations: **441**
- Average Salary: **\$95,653**

These KPIs provide a quick overview of the current data job market.

(Insert Screenshot 1: KPI Section of Dashboard)

10.2 Top Technical Skills

This visualization highlights the most frequently requested technical skills across data-related job postings.

The analysis indicates that employers consistently seek professionals with strong programming, database management, cloud computing, and analytical skills.

This information helps students understand which skills should be prioritized when preparing for careers in data analytics and data science.

(Insert Screenshot 2: Top Skills Chart)

10.3 Hiring Locations

The dashboard presents the locations with the highest concentration of data-related job opportunities.

The analysis demonstrates that data professionals are in demand across numerous regions, with several major metropolitan areas contributing significantly to job availability.

Understanding geographical demand enables job seekers to identify regions offering greater employment opportunities.

10.4 Work Type Distribution

The work type analysis categorizes positions into:

- Full-Time
- Contract
- Internship
- Part-Time
- Temporary
- Volunteer

The findings indicate that Full-Time employment represents the majority of available opportunities, while contract-based roles form the second largest category.

10.5 Highest Paying Roles

The salary analysis identifies job positions offering the highest average compensation.

Senior-level positions, Machine Learning Engineers, Principal Data Scientists, Analytics Engineers, and AI-related roles generally receive higher salaries due to their advanced technical responsibilities and experience requirements.

(Insert Screenshot 3: Salary Analysis Chart)

11. Key Findings

The project generated several valuable insights regarding the current data job market.

The most significant findings include:

- A total of **1,717** data-related job postings were identified from the original dataset.
- More than **1,018 companies** were actively recruiting data professionals.
- Job opportunities were distributed across **441 different locations**.
- The calculated average salary for analyzed data roles was approximately **\$95,653**.
- Full-Time employment dominated the market, indicating strong demand for permanent positions.
- Senior-level data positions generally offered significantly higher salaries than entry-level roles.
- Technical skills such as Python, SQL, cloud platforms, and analytical tools appeared frequently within job descriptions, emphasizing their importance for aspiring data professionals.

These findings provide practical guidance for students, job seekers, and professionals planning careers in data analytics.

12. Challenges Faced

Several challenges were encountered during the development of this project.

The dataset consisted of multiple CSV files requiring careful exploration before analysis could begin.

Some tables contained broad skill categories instead of detailed technical skills, requiring additional investigation to determine the most appropriate source of information.

Filtering only data-related job postings while maintaining data quality required multiple iterations of keyword selection and validation.

Finally, designing a clear and informative Tableau dashboard involved selecting appropriate visualizations that effectively communicated business insights.

These challenges provided valuable practical experience in handling real-world datasets.

13. Skills Learned

This project strengthened several technical and analytical skills.

Technical Skills

- Python Programming
- Pandas
- Data Cleaning
- Data Preprocessing
- Exploratory Data Analysis (EDA)
- Tableau Dashboard Development
- Data Visualization
- CSV Data Handling

Analytical Skills

- Business Problem Solving
- Data Interpretation
- Dashboard Design
- Trend Analysis
- Data-Driven Decision Making

Professional Skills

- Documentation
 - Project Planning
 - Data Presentation
 - Technical Reporting
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14. Conclusion

The Data Jobs Market Analysis Dashboard successfully transformed a large collection of LinkedIn job postings into meaningful business insights.

Using Python for data analysis and Tableau for visualization, the project identified hiring trends, salary patterns, employment types, hiring locations, and required technical skills within the data industry.

The dashboard provides an intuitive overview of the current job market and demonstrates how analytical techniques can support career planning and organizational decision-making.

Beyond the technical implementation, the project strengthened practical skills in data analysis, business intelligence, dashboard development, and technical communication.

Overall, the project demonstrates the complete workflow of a modern data analytics project—from raw data to actionable insights.

15. Future Scope

Although the project achieved its objectives, several improvements can be implemented in future versions.

Possible enhancements include:

- Integrating real-time job posting data using web scraping or APIs.
- Developing machine learning models to predict salaries based on skills and experience.
- Creating a web-based interactive dashboard using Streamlit or Power BI Service.
- Implementing AI-powered career recommendation systems.
- Performing sentiment analysis on job descriptions.
- Comparing hiring trends across different countries and industries.

These enhancements would further increase the practical value of the project.

16. References

1. LinkedIn Job Postings Dataset (2023–2024)
2. Python Documentation — <https://docs.python.org/>
3. Pandas Documentation — <https://pandas.pydata.org/>
4. Matplotlib Documentation — <https://matplotlib.org/>

5. Tableau Documentation — <https://www.tableau.com/learn>
 6. Jupyter Notebook Documentation — <https://jupyter.org/>
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